Full Stack Engineering

Project Synopsis

FSEG09-18

Semester-VI (Batch-2023)

Social Media Platform

Supervised By:

Md Nurul Islam

Submitted By:

Jatin Dhull(2310990701)
Navjot Singh(2310990748)
Ashish(2310991371)

Department of Computer Science and Engineering ChitkaraUniversity Institute of Engineering & Technology, Chitkara University, Punjab

1. Introduction

A Social Media Platform is a web-based application that enables users to connect, communicate, and share content digitally. In today's world, social networking plays a vital role in communication, content sharing, and community building.

The proposed **TrueVoice (**Social Media Platform) allows users to create accounts, share posts (text, images, videos), interact through likes and comments, and follow other users. It also provides real-time notifications and a personalized news feed.

This project will be developed using **React.js for frontend**, **Node.js and Express.js for backend**, and **MySQL for database management**. The system will follow a RESTful API architecture with secure authentication using JWT.

2. Problem Statement

Many existing academic-level social media platforms lack real-world features and proper system design. Common issues include:

- Weak or no authentication mechanisms
- Lack of proper authorization controls
- Poor database structure
- No real-time interaction features
- Limited scalability and responsiveness

Therefore, there is a need to develop a **production-level platform like TrueVoice** that includes secure authentication, proper authorization, efficient database design, and interactive user features.

3. Objectives of the Project

The main objectives of the **TrueVoice** are:

- To allow users to create and share multimedia posts
- To implement like, comment, and follow functionalities
- To provide a personalized news feed for users
- To enable real-time notifications for user interactions
- To implement authorization so only owners can modify their content
- To design a normalized relational database
- To build a responsive and user-friendly interface

4. Scope of the Project

The **TrueVoice platform** will allow users to connect and interact in a secure and scalable environment. It will support user-generated content, social interactions, and notifications.

The system will provide APIs for managing users, posts, comments, likes, follows, and notifications. All data will be stored in a **MySQL relational database**.

The project can be extended in the future with features such as:

- Real-time chat system
- Video streaming
- Advanced recommendation algorithms
- Mobile application integration

5. Proposed System

The proposed system will automate the process of task management and project tracking. Users will be able to create projects, assign tasks, track activities, and collaborate through comments.

The system will also support task boards where tasks can be organized into different stages such as **To Do, In Progress, and Completed**. Notifications will inform users about updates, while activity logs will maintain a record of actions performed within the system.

6. Technology Stack

The following technologies will be used:

- **React.js** – Frontend development
- **Node.js** – Backend runtime environment
- **Express.js** – Web framework for APIs
- **MySQL** – Relational database
- **JWT (JSON Web Tokens)** – Authentication
- **bcrypt** – Password hashing
- **Multer + Cloudinary** – File upload and storage
- **Nodemailer** – Email verification service
- **Axios** – HTTP client
- **CSS Modules / Tailwind CSS** – Styling
- **Git & GitHub** – Version control

7. System Features

User Authentication

- User registration with email verification
- Secure login using JWT

User Profiles

- View and edit profile details
- Display posts and follower/following counts

Post Management

- Create, update, and delete posts
- Upload images
- View post interactions

Social Interactions

- Like and comment on posts
- Follow/unfollow users
- Personalized news feed

Notifications

- Get notified for likes, comments, and followers
- View and manage notifications

8. Authorization and Roles

User

- Create and manage own posts
- Like, comment, and follow users
- View feed and notifications

System Authorization Rules

- Only post owners can edit or delete their posts
- Protected routes require valid JWT tokens

9. Expected Outcome

The **TrueVoice Social Media Platform** will provide a fully functional environment where users can interact seamlessly. It will ensure secure authentication, efficient data handling, and responsive design.

The platform will enhance user engagement and demonstrate real-world full-stack development skills.

10. Conclusion

The **TrueVoice** demonstrates a complete understanding of modern web development by integrating frontend, backend, database, and security concepts.

By using technologies such as React.js, Node.js, Express.js, and MySQL, the system ensures scalability, security, and performance. This project serves as a strong foundation for building real-world social networking applications.